

COMP4028 – Data Analysis using R Programming

Food Allergy Analysis – Zenodo Dataset

Group 6

28 March 2026‘

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1 Introduction

This report performs a comprehensive data analysis on the **Food Allergy Analysis Dataset** sourced from Zenodo. The dataset contains de-identified patient records tracking food allergy diagnoses, atopic conditions, and demographic information for over 333,000 subjects. The analysis is performed using R Programming and covers data exploration, manipulation, statistical analysis, and visualization.

2 Load Required Libraries

```
library(dplyr)
library(ggplot2)
library(tidyr)
library(reshape2)
```

3 Load the Dataset

```
# Read the gzipped CSV file directly
df <- read.csv("C:/Users/kvnsc/Downloads/food-allergy-analysis-Zenodo.csv", stringsAsFactors = FALSE)
```

4 1. Print the Structure of the Dataset

```
str(df)
```

```

## 'data.frame':    333200 obs. of  50 variables:
## $ SUBJECT_ID      : int  1 2 3 4 5 6 7 8 9 10 ...
## $ BIRTH_YEAR      : int  2006 1994 2006 2004 2006 2006 2006 2006 2006 2006 ...
## $ GENDER_FACTOR    : chr   "S1 - Female" "S1 - Female" "S0 - Male" "S0 - Male" ...
## $ RACE_FACTOR      : chr   "R1 - Black" "R0 - White" "R0 - White" "R4 - Unknown" ...
## $ ETHNICITY_FACTOR : chr   "E0 - Non-Hispanic" "E0 - Non-Hispanic" "E1 - Hispanic" "E1 - Hispanic" ...
## $ PAYER_FACTOR     : chr   "P1 - Medicaid" "P0 - Non-Medicaid" "P0 - Non-Medicaid" "P0 - Non-Medicaid" ...
## $ ATOPIC_MARCH_COHORT : logi  FALSE FALSE TRUE FALSE FALSE FALSE ...
## $ AGE_START_YEARS  : num   0.0931 12.2327 0.011 2.3984 0.0137 ...
## $ AGE_END_YEARS    : num   3.16 18.88 6.73 9.11 6.19 ...
## $ SHELLFISH_ALG_START : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ SHELLFISH_ALG_END   : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ FISH_ALG_START     : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ FISH_ALG_END       : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ MILK_ALG_START     : num   NA NA 1 NA NA ...
## $ MILK_ALG_END       : num   NA NA 1 NA NA ...
## $ SOY_ALG_START      : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ SOY_ALG_END        : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ EGG_ALG_START      : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ EGG_ALG_END        : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ WHEAT_ALG_START    : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ WHEAT_ALG_END      : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ PEANUT_ALG_START   : num   NA NA NA NA NA ...
## $ PEANUT_ALG_END     : num   NA NA NA NA NA ...
## $ SESAME_ALG_START   : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ SESAME_ALG_END     : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ TREENUT_ALG_START  : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ TREENUT_ALG_END    : logi  NA NA NA NA NA NA ...
## $ WALNUT_ALG_START   : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ WALNUT_ALG_END     : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ PECAN_ALG_START    : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ PECAN_ALG_END      : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ PISTACH_ALG_START  : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ PISTACH_ALG_END    : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ ALMOND_ALG_START   : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ ALMOND_ALG_END     : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ BRAZIL_ALG_START   : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ BRAZIL_ALG_END     : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ HAZELNUT_ALG_START : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ HAZELNUT_ALG_END   : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ CASHEW_ALG_START   : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ CASHEW_ALG_END     : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ ATOPIC_DERM_START  : num   NA NA 4.88 NA NA ...
## $ ATOPIC_DERM_END    : num   NA NA NA NA NA ...
## $ ALLERGIC_RHINITIS_START : num  NA NA 3.92 NA NA ...
## $ ALLERGIC_RHINITIS_END : num  NA NA 6.16 NA NA ...
## $ ASTHMA_START       : num   NA NA 5.13 NA NA ...
## $ ASTHMA_END         : num   NA NA NA NA NA NA NA NA NA NA NA ...
## $ FIRST_ASTHMARX     : num   NA 12.3 1.4 NA NA ...
## $ LAST_ASTHMARX      : num   NA 18.88 6.16 NA NA ...
## $ NUM_ASTHMARX       : int   NA 2 4 NA NA NA NA NA NA NA ...

```

5 2. List the Variables in the Dataset

```
names(df)
```

```
## [1] "SUBJECT_ID"          "BIRTH_YEAR"
## [3] "GENDER_FACTOR"       "RACE_FACTOR"
## [5] "ETHNICITY_FACTOR"    "PAYER_FACTOR"
## [7] "ATOPIC_MARCH_COHORT" "AGE_START_YEARS"
## [9] "AGE_END_YEARS"       "SHELLFISH_ALG_START"
## [11] "SHELLFISH_ALG_END"   "FISH_ALG_START"
## [13] "FISH_ALG_END"        "MILK_ALG_START"
## [15] "MILK_ALG_END"        "SOY_ALG_START"
## [17] "SOY_ALG_END"         "EGG_ALG_START"
## [19] "EGG_ALG_END"         "WHEAT_ALG_START"
## [21] "WHEAT_ALG_END"       "PEANUT_ALG_START"
## [23] "PEANUT_ALG_END"      "SESAME_ALG_START"
## [25] "SESAME_ALG_END"      "TREENUT_ALG_START"
## [27] "TREENUT_ALG_END"     "WALNUT_ALG_START"
## [29] "WALNUT_ALG_END"      "PECAN_ALG_START"
## [31] "PECAN_ALG_END"       "PISTACH_ALG_START"
## [33] "PISTACH_ALG_END"     "ALMOND_ALG_START"
## [35] "ALMOND_ALG_END"      "BRAZIL_ALG_START"
## [37] "BRAZIL_ALG_END"      "HAZELNUT_ALG_START"
## [39] "HAZELNUT_ALG_END"    "CASHEW_ALG_START"
## [41] "CASHEW_ALG_END"      "ATOPIC_DERM_START"
## [43] "ATOPIC_DERM_END"     "ALLERGIC_RHINITIS_START"
## [45] "ALLERGIC_RHINITIS_END" "ASTHMA_START"
## [47] "ASTHMA_END"          "FIRST_ASTHMARX"
## [49] "LAST_ASTHMARX"       "NUM_ASTHMARX"
```

6 3. Print the Top 15 Rows

```
head(df, 15)
```

```
##   SUBJECT_ID BIRTH_YEAR GENDER_FACTOR RACE_FACTOR ETHNICITY_FACTOR
## 1          1      2006    S1 - Female    R1 - Black  E0 - Non-Hispanic
## 2          2      1994    S1 - Female    R0 - White  E0 - Non-Hispanic
## 3          3      2006     S0 - Male     R0 - White   E1 - Hispanic
## 4          4      2004     S0 - Male    R4 - Unknown   E1 - Hispanic
## 5          5      2006    S1 - Female    R1 - Black  E0 - Non-Hispanic
## 6          6      2006     S0 - Male     R0 - White  E0 - Non-Hispanic
## 7          7      2006    S1 - Female    R0 - White  E0 - Non-Hispanic
## 8          8      2006    S1 - Female    R1 - Black  E0 - Non-Hispanic
## 9          9      2006     S0 - Male     R0 - White  E0 - Non-Hispanic
## 10         10      2006    S1 - Female    R1 - Black  E0 - Non-Hispanic
## 11         11      2006     S0 - Male    R4 - Unknown  E0 - Non-Hispanic
## 12         12      2005     S0 - Male     R0 - White  E0 - Non-Hispanic
```

## 13	13	2006	S1 - Female	R0 - White	E0 - Non-Hispanic
## 14	14	2006	S0 - Male	R4 - Unknown	E1 - Hispanic
## 15	15	2006	S1 - Female	R0 - White	E0 - Non-Hispanic
##	PAYER_FACTOR	ATOPIC_MARCH_COHORT	AGE_START_YEARS	AGE_END_YEARS	
## 1	P1 - Medicaid	FALSE	0.09308693	3.164956	
## 2	P0 - Non-Medicaid	FALSE	12.23271732	18.880219	
## 3	P0 - Non-Medicaid	TRUE	0.01095140	6.726899	
## 4	P0 - Non-Medicaid	FALSE	2.39835729	9.111567	
## 5	P0 - Non-Medicaid	FALSE	0.01368925	6.193018	
## 6	P1 - Medicaid	FALSE	0.08761123	4.958248	
## 7	P0 - Non-Medicaid	FALSE	0.01095140	6.677618	
## 8	P0 - Non-Medicaid	TRUE	0.03285421	6.286105	
## 9	P0 - Non-Medicaid	TRUE	0.01916496	6.726899	
## 10	P1 - Medicaid	TRUE	0.02190281	6.275154	
## 11	P0 - Non-Medicaid	TRUE	0.01916496	7.003422	
## 12	P0 - Non-Medicaid	FALSE	1.89733060	5.900068	
## 13	P0 - Non-Medicaid	TRUE	0.01642710	6.663929	
## 14	P1 - Medicaid	FALSE	0.01368925	5.911020	
## 15	P0 - Non-Medicaid	FALSE	0.01368925	2.866530	
##	SHELLFISH_ALG_START	SHELLFISH_ALG_END	FISH_ALG_START	FISH_ALG_END	
## 1	NA	NA	NA	NA	
## 2	NA	NA	NA	NA	
## 3	NA	NA	NA	NA	
## 4	NA	NA	NA	NA	
## 5	NA	NA	NA	NA	
## 6	NA	NA	NA	NA	
## 7	NA	NA	NA	NA	
## 8	NA	NA	NA	NA	
## 9	NA	NA	NA	NA	
## 10	NA	NA	NA	NA	
## 11	NA	NA	NA	NA	
## 12	NA	NA	NA	NA	
## 13	NA	NA	NA	NA	
## 14	NA	NA	NA	NA	
## 15	NA	NA	NA	NA	
##	MILK_ALG_START	MILK_ALG_END	SOY_ALG_START	SOY_ALG_END	EGG_ALG_START
## 1	NA	NA	NA	NA	NA
## 2	NA	NA	NA	NA	NA
## 3	1.002053	1.002053	NA	NA	NA
## 4	NA	NA	NA	NA	NA
## 5	NA	NA	NA	NA	NA
## 6	NA	NA	NA	NA	NA
## 7	NA	NA	NA	NA	NA
## 8	NA	NA	NA	NA	NA
## 9	NA	NA	NA	NA	NA
## 10	NA	NA	NA	NA	NA
## 11	NA	NA	NA	NA	NA
## 12	NA	NA	NA	NA	NA
## 13	NA	NA	NA	NA	NA
## 14	NA	NA	NA	NA	NA
## 15	NA	NA	NA	NA	NA
##	EGG_ALG_END	WHEAT_ALG_START	WHEAT_ALG_END	PEANUT_ALG_START	PEANUT_ALG_END
## 1	NA	NA	NA	NA	NA
## 2	NA	NA	NA	NA	NA

## 3	NA	NA	NA	NA	NA
## 4	NA	NA	NA	NA	NA
## 5	NA	NA	NA	NA	NA
## 6	NA	NA	NA	NA	NA
## 7	NA	NA	NA	NA	NA
## 8	NA	NA	NA	NA	NA
## 9	NA	NA	NA	1.221081	1.221081
## 10	NA	NA	NA	NA	NA
## 11	NA	NA	NA	NA	NA
## 12	NA	NA	NA	NA	NA
## 13	NA	NA	NA	NA	NA
## 14	NA	NA	NA	NA	NA
## 15	NA	NA	NA	NA	NA
##	SESAME_ALG_START	SESAME_ALG_END	TREENUT_ALG_START	TREENUT_ALG_END	
## 1	NA	NA	NA	NA	
## 2	NA	NA	NA	NA	
## 3	NA	NA	NA	NA	
## 4	NA	NA	NA	NA	
## 5	NA	NA	NA	NA	
## 6	NA	NA	NA	NA	
## 7	NA	NA	NA	NA	
## 8	NA	NA	NA	NA	
## 9	NA	NA	NA	NA	
## 10	NA	NA	NA	NA	
## 11	NA	NA	NA	NA	
## 12	NA	NA	NA	NA	
## 13	NA	NA	NA	NA	
## 14	NA	NA	NA	NA	
## 15	NA	NA	NA	NA	
##	WALNUT_ALG_START	WALNUT_ALG_END	PECAN_ALG_START	PECAN_ALG_END	
## 1	NA	NA	NA	NA	
## 2	NA	NA	NA	NA	
## 3	NA	NA	NA	NA	
## 4	NA	NA	NA	NA	
## 5	NA	NA	NA	NA	
## 6	NA	NA	NA	NA	
## 7	NA	NA	NA	NA	
## 8	NA	NA	NA	NA	
## 9	NA	NA	NA	NA	
## 10	NA	NA	NA	NA	
## 11	NA	NA	NA	NA	
## 12	NA	NA	NA	NA	
## 13	NA	NA	NA	NA	
## 14	NA	NA	NA	NA	
## 15	NA	NA	NA	NA	
##	PISTACH_ALG_START	PISTACH_ALG_END	ALMOND_ALG_START	ALMOND_ALG_END	
## 1	NA	NA	NA	NA	
## 2	NA	NA	NA	NA	
## 3	NA	NA	NA	NA	
## 4	NA	NA	NA	NA	
## 5	NA	NA	NA	NA	
## 6	NA	NA	NA	NA	
## 7	NA	NA	NA	NA	
## 8	NA	NA	NA	NA	

## 9	NA	NA	NA	NA
## 10	NA	NA	NA	NA
## 11	NA	NA	NA	NA
## 12	NA	NA	NA	NA
## 13	NA	NA	NA	NA
## 14	NA	NA	NA	NA
## 15	NA	NA	NA	NA
##	BRAZIL_ALG_START	BRAZIL_ALG_END	HAZELNUT_ALG_START	HAZELNUT_ALG_END
## 1	NA	NA	NA	NA
## 2	NA	NA	NA	NA
## 3	NA	NA	NA	NA
## 4	NA	NA	NA	NA
## 5	NA	NA	NA	NA
## 6	NA	NA	NA	NA
## 7	NA	NA	NA	NA
## 8	NA	NA	NA	NA
## 9	NA	NA	NA	NA
## 10	NA	NA	NA	NA
## 11	NA	NA	NA	NA
## 12	NA	NA	NA	NA
## 13	NA	NA	NA	NA
## 14	NA	NA	NA	NA
## 15	NA	NA	NA	NA
##	CASHEW_ALG_START	CASHEW_ALG_END	ATOPIIC_DERM_START	ATOPIIC_DERM_END
## 1	NA	NA	NA	NA
## 2	NA	NA	NA	NA
## 3	NA	NA	4.884326	NA
## 4	NA	NA	NA	NA
## 5	NA	NA	NA	NA
## 6	NA	NA	NA	NA
## 7	NA	NA	NA	NA
## 8	NA	NA	NA	NA
## 9	NA	NA	1.015743	1.015743
## 10	NA	NA	NA	NA
## 11	NA	NA	NA	NA
## 12	NA	NA	NA	NA
## 13	NA	NA	NA	NA
## 14	NA	NA	NA	NA
## 15	NA	NA	NA	NA
##	ALLERGIC_RHINITIS_START	ALLERGIC_RHINITIS_END	ASTHMA_START	ASTHMA_END
## 1	NA	NA	NA	NA
## 2	NA	NA	NA	NA
## 3	3.917864	6.157426	5.127995	NA
## 4	NA	NA	NA	NA
## 5	NA	NA	NA	NA
## 6	NA	NA	NA	NA
## 7	NA	NA	NA	NA
## 8	NA	NA	NA	NA
## 9	NA	NA	NA	NA
## 10	NA	NA	NA	NA
## 11	NA	NA	NA	NA
## 12	NA	NA	NA	NA
## 13	NA	NA	NA	NA
## 14	NA	NA	1.002053	3.789185

		NA	NA	NA	NA
	FIRST_ASTHMARX	LAST_ASTHMARX	NUM_ASTHMARX		
## 1	NA	NA	NA		
## 2	12.262834	18.880219	2		
## 3	1.404517	6.157426	4		
## 4	NA	NA	NA		
## 5	NA	NA	NA		
## 6	NA	NA	NA		
## 7	NA	NA	NA		
## 8	NA	NA	NA		
## 9	NA	NA	NA		
## 10	NA	NA	NA		
## 11	7.003422	7.003422	1		
## 12	NA	NA	NA		
## 13	NA	NA	NA		
## 14	1.007529	1.524983	8		
## 15	NA	NA	NA		

7 4. User-Defined Function

A custom function is defined to calculate the **duration** of a condition (e.g. asthma) given a start and end age column. It returns the duration in years, or NA if either value is missing.

```
# User-defined function: calculate duration between two age columns
calculate_duration <- function(start_age, end_age) {
  duration <- ifelse(!is.na(start_age) & !is.na(end_age),
    round(end_age - start_age, 2),
    NA)
  return(duration)
}

# Apply to asthma start and end ages
df$ASTHMA_DURATION <- calculate_duration(df$ASTHMA_START, df$ASTHMA_END)

# Preview results
head(df[!is.na(df$ASTHMA_DURATION), c("SUBJECT_ID", "ASTHMA_START", "ASTHMA_END", "ASTHMA_DURATION")],
```

	SUBJECT_ID	ASTHMA_START	ASTHMA_END	ASTHMA_DURATION
## 14	14	1.002053	3.789185	2.79
## 64	64	2.748802	6.195756	3.45
## 78	78	3.290897	8.068446	4.78
## 125	125	1.577002	12.136893	10.56
## 143	143	9.552361	12.016427	2.46
## 148	148	1.002053	4.555784	3.55
## 172	172	1.905544	6.220397	4.31
## 189	189	1.453799	4.180698	2.73
## 218	218	3.104723	7.586585	4.48
## 220	220	1.826146	6.521561	4.70

8 5. Filter Rows Based on Logical Criteria

Filter to keep only patients who: - Are part of the **Atopic March Cohort** (ATOPIC_MARCH_COHORT == TRUE) - Have a recorded peanut allergy start age

```
filtered_df <- df %>%  
  filter(ATOPIC_MARCH_COHORT == TRUE & !is.na(PEANUT_ALG_START))  
  
cat("Rows after filtering:", nrow(filtered_df), "\n")
```

```
## Rows after filtering: 1153
```

```
head(filtered_df[, c("SUBJECT_ID", "GENDER_FACTOR", "RACE_FACTOR",  
  "PEANUT_ALG_START", "ATOPIC_MARCH_COHORT")], 10)
```

##	SUBJECT_ID	GENDER_FACTOR	RACE_FACTOR	PEANUT_ALG_START	ATOPIC_MARCH_COHORT
## 1	9	S0 - Male	R0 - White	1.2210815	TRUE
## 2	66	S1 - Female	R1 - Black	2.5215606	TRUE
## 3	163	S0 - Male	R0 - White	1.7330595	TRUE
## 4	198	S0 - Male	R0 - White	1.5112936	TRUE
## 5	223	S0 - Male	R4 - Unknown	1.4428474	TRUE
## 6	11908	S0 - Male	R1 - Black	1.2594114	TRUE
## 7	11933	S0 - Male	R1 - Black	0.9445585	TRUE
## 8	11984	S0 - Male	R4 - Unknown	5.2073922	TRUE
## 9	12026	S1 - Female	R1 - Black	0.7556468	TRUE
## 10	12177	S0 - Male	R0 - White	1.5140315	TRUE

9 6. Identify Dependent & Independent Variables and Reshape

Independent variables: AGE_START_YEARS, GENDER_FACTOR, RACE_FACTOR, PAYER_FACTOR

Dependent variable: NUM_ASTHMARX (number of asthma prescriptions — outcome of interest)

A new data frame is created by joining these selected variables.

```
# Select independent and dependent variables  
independent_vars <- df %>%  
  select(SUBJECT_ID, AGE_START_YEARS, GENDER_FACTOR, RACE_FACTOR, PAYER_FACTOR)  
  
dependent_var <- df %>%  
  select(SUBJECT_ID, NUM_ASTHMARX)  
  
# Join into a new data frame  
analysis_df <- inner_join(independent_vars, dependent_var, by = "SUBJECT_ID")  
  
cat("Dimensions of reshaped data frame:", dim(analysis_df), "\n")
```

```
## Dimensions of reshaped data frame: 333200 6
```

```
head(analysis_df, 10)
```

```
##      SUBJECT_ID AGE_START_YEARS GENDER_FACTOR RACE_FACTOR    PAYER_FACTOR
## 1             1      0.09308693    S1 - Female    R1 - Black    P1 - Medicaid
## 2             2     12.23271732    S1 - Female    R0 - White    P0 - Non-Medicaid
## 3             3      0.01095140     S0 - Male     R0 - White    P0 - Non-Medicaid
## 4             4      2.39835729     S0 - Male    R4 - Unknown    P0 - Non-Medicaid
## 5             5      0.01368925    S1 - Female    R1 - Black    P0 - Non-Medicaid
## 6             6      0.08761123     S0 - Male     R0 - White     P1 - Medicaid
## 7             7      0.01095140    S1 - Female    R0 - White    P0 - Non-Medicaid
## 8             8      0.03285421    S1 - Female    R1 - Black    P0 - Non-Medicaid
## 9             9      0.01916496     S0 - Male     R0 - White    P0 - Non-Medicaid
## 10            10      0.02190281    S1 - Female    R1 - Black     P1 - Medicaid
##      NUM_ASTHMARX
## 1             NA
## 2              2
## 3              4
## 4             NA
## 5             NA
## 6             NA
## 7             NA
## 8             NA
## 9             NA
## 10            NA
```

10 7. Remove Missing Values

```
cat("Rows before removing NAs:", nrow(analysis_df), "\n")
```

```
## Rows before removing NAs: 333200
```

```
analysis_df_clean <- na.omit(analysis_df)
```

```
cat("Rows after removing NAs:", nrow(analysis_df_clean), "\n")
```

```
## Rows after removing NAs: 117550
```

```
cat("Rows removed:", nrow(analysis_df) - nrow(analysis_df_clean), "\n")
```

```
## Rows removed: 215650
```

11 8. Identify and Remove Duplicated Data

```
cat("Number of duplicate rows:", sum(duplicated(df)), "\n")
```

```
## Number of duplicate rows: 0
```

```
# Remove duplicates
```

```
df_no_dupes <- df[!duplicated(df), ]
```

```
cat("Rows before deduplication:", nrow(df), "\n")
```

```
## Rows before deduplication: 333200
```

```
cat("Rows after deduplication:", nrow(df_no_dupes), "\n")
```

```
## Rows after deduplication: 333200
```

12 9. Reorder Rows in Descending Order

Reorder the cleaned analysis data frame by NUM_ASTHMARX in descending order (patients with the most asthma prescriptions first).

```
analysis_df_sorted <- analysis_df_clean %>%  
  arrange(desc(NUM_ASTHMARX), desc(AGE_START_YEARS))  
  
head(analysis_df_sorted, 10)
```

```
##      SUBJECT_ID AGE_START_YEARS GENDER_FACTOR      RACE_FACTOR  
## 1      330932      6.87748118      S0 - Male R2 - Asian or Pacific Islander  
## 2      157238      3.12662560      S1 - Female      R1 - Black  
## 3      216722      8.23271732      S1 - Female      R1 - Black  
## 4      284583      0.03285421      S1 - Female      R1 - Black  
## 5         5149      4.88980151      S0 - Male      R0 - White  
## 6       33425      3.04175222      S1 - Female      R4 - Unknown  
## 7      255373      0.48733744      S1 - Female      R0 - White  
## 8       77582      0.29021218      S1 - Female      R1 - Black  
## 9      157126      2.38740589      S0 - Male      R1 - Black  
## 10     37012      1.95756331      S0 - Male      R1 - Black  
##      PAYER_FACTOR NUM_ASTHMARX  
## 1 P1 - Medicaid      118  
## 2 P1 - Medicaid      118  
## 3 P1 - Medicaid      110  
## 4 P1 - Medicaid      109  
## 5 P1 - Medicaid      106  
## 6 P1 - Medicaid      100  
## 7 P1 - Medicaid       98  
## 8 P1 - Medicaid       97  
## 9 P1 - Medicaid       94  
## 10 P1 - Medicaid       94
```

13 10. Rename Column Names

```
analysis_df_renamed <- analysis_df_clean %>%
  rename(
    Patient_ID      = SUBJECT_ID,
    Age_At_Start    = AGE_START_YEARS,
    Gender          = GENDER_FACTOR,
    Race            = RACE_FACTOR,
    Insurance_Type   = PAYER_FACTOR,
    Asthma_Rx_Count = NUM_ASTHMARX
  )

names(analysis_df_renamed)
```

```
## [1] "Patient_ID"      "Age_At_Start"    "Gender"          "Race"
## [5] "Insurance_Type"  "Asthma_Rx_Count"
```

```
head(analysis_df_renamed, 5)
```

```
##   Patient_ID Age_At_Start      Gender      Race Insurance_Type
## 2           2  12.23271732 S1 - Female  R0 - White P0 - Non-Medicaid
## 3           3   0.01095140 S0 - Male   R0 - White P0 - Non-Medicaid
## 11          11   0.01916496 S0 - Male   R4 - Unknown P0 - Non-Medicaid
## 14          14   0.01368925 S0 - Male   R4 - Unknown P1 - Medicaid
## 19          19   0.01642710 S0 - Male   R0 - White P0 - Non-Medicaid
##   Asthma_Rx_Count
## 2                 2
## 3                 4
## 11                1
## 14                8
## 19               10
```

14 11. Add New Variables Using a Mathematical Function

Add two new computed columns: - `Asthma_Rx_Double`: number of asthma prescriptions multiplied by 2 -
`Age_At_Start_Months`: age at start converted from years to months

```
analysis_df_renamed <- analysis_df_renamed %>%
  mutate(
    Asthma_Rx_Double = Asthma_Rx_Count * 2,
    Age_At_Start_Months = round(Age_At_Start * 12, 1)
  )

head(analysis_df_renamed[, c("Patient_ID", "Age_At_Start", "Age_At_Start_Months",
                             "Asthma_Rx_Count", "Asthma_Rx_Double")], 10)
```

	Patient_ID	Age_At_Start	Age_At_Start_Months	Asthma_Rx_Count	Asthma_Rx_Double
## 2	2	12.23271732	146.8	2	4
## 3	3	0.01095140	0.1	4	8
## 11	11	0.01916496	0.2	1	2
## 14	14	0.01368925	0.2	8	16
## 19	19	0.01642710	0.2	10	20
## 21	21	0.01368925	0.2	1	2
## 22	22	0.01368925	0.2	15	30
## 23	23	3.28815880	39.5	15	30
## 24	24	7.78370979	93.4	4	8
## 30	30	2.50787132	30.1	3	6

15 12. Create a Training Set Using Random Number Generator

```
# Set seed for reproducibility
set.seed(42)

# Generate a random 70% training set
train_index <- sample(1:nrow(analysis_df_renamed),
                      size = floor(0.70 * nrow(analysis_df_renamed)),
                      replace = FALSE)

train_set <- analysis_df_renamed[train_index, ]
test_set  <- analysis_df_renamed[-train_index, ]

cat("Total rows:", nrow(analysis_df_renamed), "\n")
```

```
## Total rows: 117550
```

```
cat("Training set rows (70%):", nrow(train_set), "\n")
```

```
## Training set rows (70%): 82285
```

```
cat("Test set rows (30%):", nrow(test_set), "\n")
```

```
## Test set rows (30%): 35265
```

```
head(train_set, 5)
```

	Patient_ID	Age_At_Start	Gender	Race	Insurance_Type
## 183402	183402	8.72826831	S0 - Male R1 - Black	P1 - Medicaid	
## 163890	163890	6.61738535	S1 - Female R1 - Black	P0 - Non-Medicaid	
## 286044	286044	0.01916496	S1 - Female R0 - White	P0 - Non-Medicaid	
## 220784	220784	0.01368925	S0 - Male R0 - White	P0 - Non-Medicaid	
## 141572	141572	4.00273785	S0 - Male R0 - White	P0 - Non-Medicaid	
##	Asthma_Rx_Count	Asthma_Rx_Double	Age_At_Start_Months		

## 183402	21	42	104.7
## 163890	18	36	79.4
## 286044	1	2	0.2
## 220784	5	10	0.2
## 141572	1	2	48.0

16 13. Print Summary Statistics

```
summary(analysis_df_renamed)
```

```
##      Patient_ID      Age_At_Start      Gender      Race
## Min.      :      2  Min.      :-1.6454  Length:117550  Length:117550
## 1st Qu.: 80224  1st Qu.: 0.0219  Class :character  Class :character
## Median :175955  Median : 1.2430  Mode  :character  Mode  :character
## Mean   :170264  Mean   : 3.3289
## 3rd Qu.:256271  3rd Qu.: 5.8316
## Max.   :333198  Max.   :17.9247
## Insurance_Type  Asthma_Rx_Count  Asthma_Rx_Double  Age_At_Start_Months
## Length:117550  Min.      : 1.000  Min.      : 2.0  Min.      :-19.70
## Class :character  1st Qu.: 1.000  1st Qu.: 2.0  1st Qu.: 0.30
## Mode  :character  Median : 3.000  Median : 6.0  Median : 14.90
##                      Mean   : 5.298  Mean   : 10.6  Mean   : 39.95
##                      3rd Qu.: 7.000  3rd Qu.: 14.0  3rd Qu.: 70.00
##                      Max.   :118.000  Max.   :236.0  Max.   :215.10
```

17 14. Statistical Functions on Numerical Variables

Using AGE_START_YEARS (age at which the patient entered the study) and NUM_ASTHMARX (number of asthma prescriptions).

```
# --- AGE_START_YEARS ---
age_values <- analysis_df_renamed$Age_At_Start

cat("=== Age At Start (Years) ===\n")
```

```
## === Age At Start (Years) ===
```

```
cat("Mean: ", round(mean(age_values), 4), "\n")
```

```
## Mean: 3.3289
```

```
cat("Median:", round(median(age_values), 4), "\n")
```

```
## Median: 1.243
```

```
cat("Range: ", round(range(age_values), 4), "\n")
```

```
## Range: -1.6454 17.9247
```

```
# Mode (custom function - R has no built-in mode for continuous data)
```

```
get_mode <- function(x) {  
  x_clean <- x[!is.na(x)]  
  unique_vals <- unique(x_clean)  
  unique_vals[which.max(tabulate(match(x_clean, unique_vals)))]  
}
```

```
cat("Mode: ", round(get_mode(age_values), 4), "\n")
```

```
## Mode: 0.011
```

```
# --- NUM_ASTHMAXX ---
```

```
rx_values <- analysis_df_renamed$Asthma_Rx_Count
```

```
cat("\n=== Number of Asthma Prescriptions ===\n")
```

```
##
```

```
## === Number of Asthma Prescriptions ===
```

```
cat("Mean: ", round(mean(rx_values), 4), "\n")
```

```
## Mean: 5.2985
```

```
cat("Median:", round(median(rx_values), 4), "\n")
```

```
## Median: 3
```

```
cat("Range: ", range(rx_values), "\n")
```

```
## Range: 1 118
```

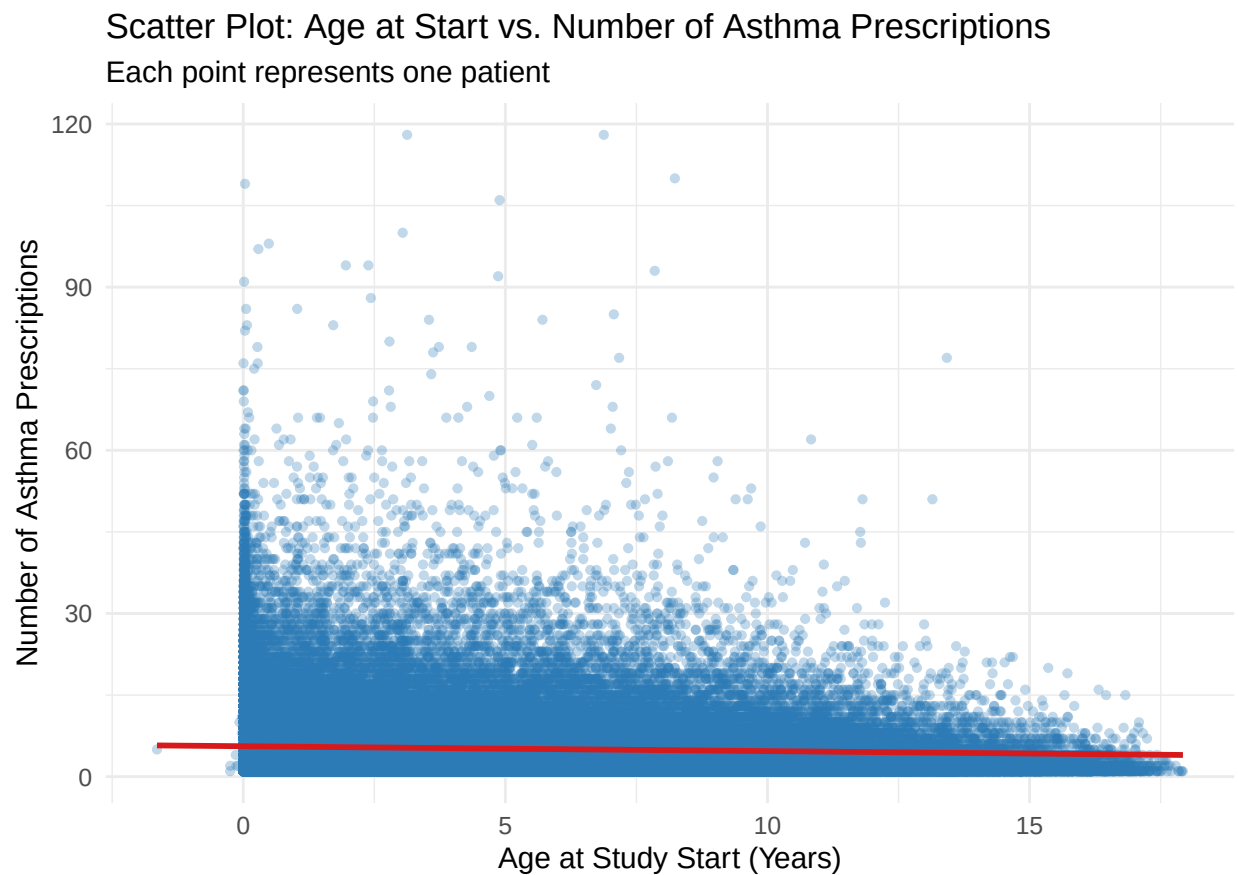
```
cat("Mode: ", get_mode(rx_values), "\n")
```

```
## Mode: 1
```

18 15. Scatter Plot

Age at Start vs. Number of Asthma Prescriptions — to explore whether older patients at study entry tend to have more asthma prescriptions.

```
ggplot(analysis_df_renamed, aes(x = Age_At_Start, y = Asthma_Rx_Count)) +
  geom_point(alpha = 0.3, color = "#2C7BB6", size = 1.2) +
  geom_smooth(method = "lm", color = "#D7191C", se = TRUE) +
  labs(
    title = "Scatter Plot: Age at Start vs. Number of Asthma Prescriptions",
    subtitle = "Each point represents one patient",
    x = "Age at Study Start (Years)",
    y = "Number of Asthma Prescriptions"
  ) +
  theme_minimal(base_size = 12)
```



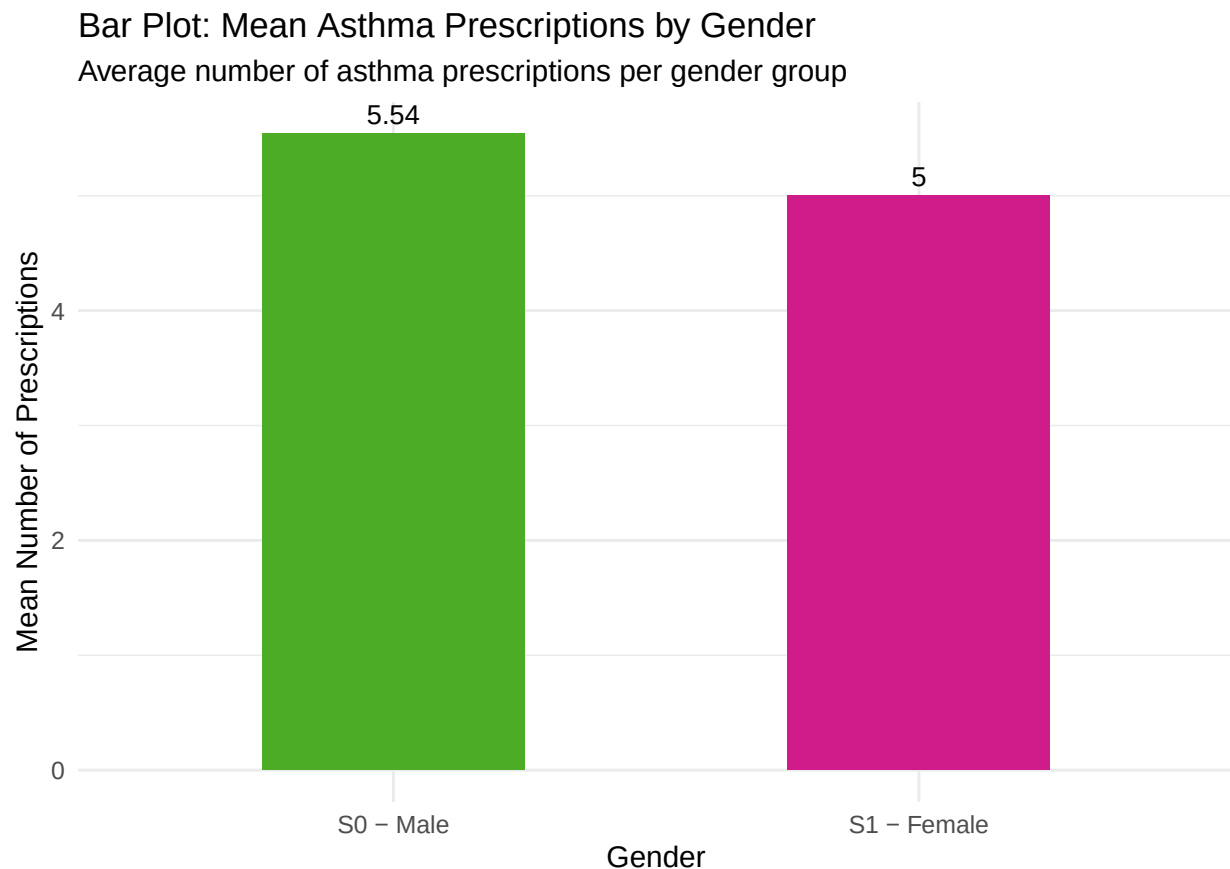
19 16. Bar Plot

Distribution of Asthma Prescription Counts by Gender — comparing the number of asthma prescriptions across male and female patients.

```
bar_data <- analysis_df_renamed %>%
  group_by(Gender) %>%
  summarise(Mean_Asthma_Rx = round(mean(Asthma_Rx_Count), 2))
```



```
ggplot(bar_data, aes(x = Gender, y = Mean_Asthma_Rx, fill = Gender)) +
  geom_bar(stat = "identity", width = 0.5, show.legend = FALSE) +
  geom_text(aes(label = Mean_Asthma_Rx), vjust = -0.5, size = 4) +
  scale_fill_manual(values = c("#4DAC26", "#D01C8B")) +
  labs(
    title = "Bar Plot: Mean Asthma Prescriptions by Gender",
    subtitle = "Average number of asthma prescriptions per gender group",
    x = "Gender",
    y = "Mean Number of Prescriptions"
  ) +
  theme_minimal(base_size = 12)
```



20 17. Pearson Correlation

Finding the correlation between **Age at Start** and **Number of Asthma Prescriptions**.

```
pearson_result <- cor.test(
  analysis_df_renamed$Age_At_Start,
  analysis_df_renamed$Asthma_Rx_Count,
  method = "pearson"
```

```

)

cat("Pearson Correlation Coefficient (r):", round(pearson_result$estimate, 4), "\n")

## Pearson Correlation Coefficient (r): -0.0574

cat("P-value:", format(pearson_result$p.value, scientific = TRUE), "\n")

## P-value: 1.952689e-86

cat("95% Confidence Interval:", round(pearson_result$conf.int, 4), "\n")

## 95% Confidence Interval: -0.0631 -0.0517

if (abs(pearson_result$estimate) < 0.2) {
  cat("Interpretation: Very weak correlation\n")
} else if (abs(pearson_result$estimate) < 0.4) {
  cat("Interpretation: Weak correlation\n")
} else if (abs(pearson_result$estimate) < 0.6) {
  cat("Interpretation: Moderate correlation\n")
} else {
  cat("Interpretation: Strong correlation\n")
}

## Interpretation: Very weak correlation

```

21 Conclusion

This analysis explored the Food Allergy dataset from Zenodo, which contains over 333,000 de-identified patient records. Key steps included:

- **Data exploration:** Reviewed structure, variables, and top rows to understand the dataset.
- **Data cleaning:** Removed missing values and duplicate records to ensure data integrity.
- **Feature engineering:** Created new variables such as asthma duration and prescription count doubled.
- **Statistical analysis:** Computed mean, median, mode, and range for key numerical variables.
- **Visualization:** Scatter and bar plots revealed patterns in asthma prescriptions by age and gender.
- **Correlation:** Pearson correlation assessed the linear relationship between age at study entry and asthma prescription frequency.

GitHub Repository: [Add your GitHub link here]